

Multi-dimensional graph linear canonical transform and its application

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ABSTRACT

Processing multi-dimensional (mD) graph data is crucial in fields such as social networks, communication networks, image processing, and signal processing due to its effective representation of complex relationships and network structures. Designing a transform method for processing these mD graph signals in the graph linear canonical domain remains a key challenge in graph signal processing. This article investigates new transforms for mD graph signals defined on Cartesian product graphs, including two-dimensional graph linear canonical transforms (2D GLCTs) based on adjacency matrices and graph Laplacian matrices. Furthermore, these transforms are extended to mD GLCTs, enabling the handling of more complex mD graph data. To demonstrate the practicality of the proposed method, this paper uses the 2D GLCT based on the Laplacian matrix as an example to detail its application in data compression.

1. Introduction

The representation and processing of data are critical research areas in the contemporary digital era. Graph data structures are particularly effective in representing complex relationships and network structures, leading to their wide application across various fields such as social networks, communication networks, image processing, and signal processing [1,2]. Recently, the analysis of graph signals has gained substantial attention, prompting the development of graph-based transforms, including the graph Fourier transform (GFT) [3–5], graph filters [6,7], and techniques for graph sampling and recovery [8,9]. Analogous to classical signal processing, a primary goal in graph signal processing is to design transformation methods that exploit the structure of signals on graphs [10–17]. Among these methodologies, the GFT has been extensively studied and remains a central topic of research.

Current research indicates that the GFT is primarily defined using adjacency matrices [3] and graph Laplacian matrices [18]. However, similar to the classical Fourier transform, a single GFT method may not fully capture the intricate structures of graph signals in certain applications. To address these limitations, researchers have developed various new transformation methods, including the spectral graph wavelet transform (SGWT) [10,19], windowed GFT (WGFT) [11,20,21], graph fractional Fourier transform (GFrFT) [12,13], windowed GFrFT (WGFrFT) [14,22], and the graph linear canonical transform (GLCT) [15]. With the continuous increase in data size and complexity, tradi-

tional one-dimensional graph transform methods fall short in effectively processing multi-dimensional (mD) graph data. Consequently, the study of mD transforms becomes crucial. These methods are capable of handling complex high-dimensional graph data, thereby better capturing the mD features inherent in graph data. This enhances both the accuracy and efficiency of data analysis and processing. Specifically, the mD GFT [16,17,23,24] and mD GFrFT [25] have been introduced to overcome the limitations of GFT in representing the spectrum of mD graph signals defined on Cartesian products of graphs.

The GLCT was first presented by Zhang [15,26] and subsequently refined by Chen [27]. The parameterization of GLCT offers considerable flexibility, leading to enhanced performance [28,29]. However, similar to the GFT, GLCT encounters challenges in accurately representing the spectrum of mD graph signals, as it does not account for the product structures within graphs [16,25]. To address these limitations, this paper first introduces the two-dimensional (2D) GLCT based on both adjacency and Laplacian matrices. This novel approach creates a transform framework specifically for 2D graph data by utilizing these matrices. Furthermore, we extend this methodology to the mD GLCT, thereby enabling efficient processing and analysis of complex mD graph data.

The study of mD GLCT holds significant practical importance. For example, in image processing, it can greatly enhance performance in tasks such as image compression, denoising, and feature extraction. In social network analysis, mD GLCT enables a more precise capture of complex relationships and behavioral patterns among users. In bioinformatics, it

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can be applied to process and analyze gene expression data, unveiling intricate interactions within biological systems. To illustrate the practical utility of the proposed methods, this paper details the application of the 2D GLCT based on Laplacian matrices for data compression. Experimental results demonstrate its effectiveness in improving data compression efficiency, offering a novel solution for the efficient storage and transmission of graph data.

The organization of this paper is as follows: Section 2 provides an overview of GFT on graphs and the basic methods of GLCT, along with the definition of Cartesian product graphs. Section 3.1 delves into the formulation of the 2D GLCT utilizing adjacency and Laplacian matrices. This is further expanded in Section 3.2, where the methodology is extended to mD GLCT. In Section 4, the application of 2D GLCT in data compression is examined, showcasing its practical utility. Finally, Section 5 provides the conclusion of the paper, summarizing the findings and potential implications.

2. Preliminaries

2.1. Graph Fourier transform

Consider a graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}, \mathbf{W}\}$, where $\mathcal{V} = \{v_0, v_1, \dots, v_{N-1}\}$ with $|\mathcal{V}| = N$ represents a finite set of vertices, $\mathcal{E} = \{(i, j) | i, j \in \mathcal{V}, j \sim i\} \subseteq \mathcal{V} \times \mathcal{V}$ denotes the set of edges, and $\mathbf{W}$ represents the weighted adjacency matrix. In this graph, if there's an edge $e = (i, j)$ connecting vertices i and j , we denote the weight of the edge as $w_{i,j}$; otherwise, $w_{i,j} = 0$ [18]. A signal on the graph $\mathcal{G}$ can be described as follows [14]:

$$f: \mathcal{V} \rightarrow \mathbb{R}$$

$$v_i \mapsto f(i). \quad (1)$$

Alternatively, we can represent $\mathbf{f}$ as a real-valued vector $\mathbf{f} = [f(0), f(1), \dots, f(N-1)]^T \in \mathbb{R}^N$, where $f(i)$ denotes the value associated with the i -th vertex.

2.1.1. Adjacency-based graph Fourier transform

The Jordan decomposition of $\mathbf{W}$ is expressed as

$$\mathbf{W} = \mathbf{V}\mathbf{J}\mathbf{V}^{-1}, \quad (2)$$

where $\mathbf{V}$ represents the matrix of generalized eigenvectors, and $\mathbf{J}$ is the matrix of Jordan blocks corresponding to the eigenvalues.

The adjacency-based graph Fourier transform (A-GFT) of a graph signal $\mathbf{f}$ is defined as [3]

$$\hat{\mathbf{f}}^{(A)} = \mathcal{F}^{(A)} \mathbf{f} = \mathbf{V}^{-1} \mathbf{f}, \quad (3)$$

where $\mathcal{F}^{(A)} = \mathbf{V}^{-1}$ is the A-GFT matrix. The inverse A-GFT is given by

$$\mathbf{f} = (\mathcal{F}^{(A)})^H \hat{\mathbf{f}}^{(A)}. \quad (4)$$

2.1.2. Laplacian-based graph Fourier transform

Assume the graph is undirected with positive edge weights. The non-normalized graph Laplacian is a symmetric difference operator, defined as $\mathcal{L} = \mathbf{D} - \mathbf{W}$, where $\mathbf{D}$ is the diagonal degree matrix of the graph $\mathcal{G}$. The i -th diagonal element d_i of $\mathbf{D}$ is equal to the sum of the weights of all the edges incident to vertex i . Since the matrix $\mathcal{L}$ is real and symmetric, it has a complete set of orthonormal eigenvectors, denoted by $\{\mathbf{u}_k\}_{k=0,1,\dots,N-1}$. These eigenvectors are ordered in ascending order according to their corresponding eigenvalues $0 = \tilde{\lambda}_0 < \tilde{\lambda}_1 \leq \tilde{\lambda}_2 \leq \dots \leq \tilde{\lambda}_{N-1}$. Thus, using the unitary column matrix $\mathbf{U} = [\mathbf{u}_0, \mathbf{u}_1, \dots, \mathbf{u}_{N-1}]$ and the diagonal matrix $\mathbf{\Lambda} = \text{diag}([\tilde{\lambda}_0, \tilde{\lambda}_1, \dots, \tilde{\lambda}_{N-1}])$, we can express

$$\mathcal{L} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^H, \quad (5)$$

where the superscript H denotes the Hermitian transpose.

The laplacian-based graph Fourier transform (L-GFT) of $\mathbf{f}$ is defined as [18]

$$\hat{f}^{(L)}(k) = \langle f, \mathbf{u}_k \rangle = \sum_{n=0}^{N-1} f(n) \mathbf{u}_k^*(n), \quad k = 0, 1, \dots, N-1. \quad (6)$$

The inverse L-GFT is given by

$$f(n) = \sum_{k=0}^{N-1} \hat{f}^{(L)}(k) \mathbf{u}_k(n), \quad n = 0, 1, \dots, N-1. \quad (7)$$

The L-GFT can be represented in vector form as $\hat{\mathbf{f}}^{(L)} = \mathcal{F}^{(L)} \mathbf{f} = \mathbf{U}^{-1} \mathbf{f}$ [5]. Similarly, the inverse L-GFT is defined as $\mathbf{f} = \mathbf{U} \hat{\mathbf{f}}^{(L)}$.

2.2. Linear canonical transform

The linear canonical transform (LCT) of the signal $f(t) \in L^2(\mathbb{R})$ with the parameters $\mathbf{M} = (a, b, c, d)$, where $ad - bc = 1$, is defined as follows [28,30]

$$L^{\mathbf{M}}\{f\}(u) = \begin{cases} \int_{-\infty}^{\infty} f(t) K_{\mathbf{M}}(t, u) dt & b \neq 0 \\ \sqrt{d} e^{i(cd/2)u^2} f(dt) & b = 0 \end{cases}, \quad (8)$$

$$K_{\mathbf{M}}(t, u) = \sqrt{\frac{1}{i2\pi b}} e^{i\left(\frac{d}{b}u^2 - \frac{2}{b}ut + \frac{a}{b}t^2\right)}, \quad (9)$$

where i represents an imaginary unit. When the parameter $\mathbf{M} = (\cos\alpha, \sin\alpha, -\sin\alpha, \cos\alpha)$ is employed, the LCT transforms into the α -th order fractional Fourier transform (FrFT).

The LCT with parameter $\mathbf{M} = (a, b, c, d)$ can be decomposed into a combination of specialized transforms by breaking down the parameter matrix. The LCT can be implemented step by step through matrix factorization [29]

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \xi & 1 \end{bmatrix} \begin{bmatrix} \sigma & 0 \\ 0 & \sigma^{-1} \end{bmatrix} \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}, \quad (10)$$

where $\mathbf{M}$ is decomposed into FrFT, scaling transform, and chirp modulation (CM), and $\xi = \frac{ac+bd}{a^2+b^2}$ represents the CM parameter, $\sigma = \sqrt{a^2+b^2}$ denotes the scaling parameter, $\alpha = \cos^{-1}\left(\frac{a}{\sigma}\right) = \sin^{-1}\left(\frac{b}{\sigma}\right)$ is the FrFT parameter.

2.3. Graph linear canonical transform

2.3.1. Adjacency-based graph linear canonical transform

The A-GFT matrix $\mathcal{F}^{(A)}$ can be decomposed as $\mathcal{F}^{(A)} = \mathbf{V}^{-1} = \mathbf{Q}_A \mathbf{R}_A \mathbf{Q}_A^H$. The adjacency-based GFrFT (A-GFrFT) of $\mathbf{f}$ is defined as [12]

$$\hat{\mathbf{f}}_{\alpha}^{(A)} = \mathcal{F}_{\alpha}^{(A)} \mathbf{f} = \mathbf{Q}_A \mathcal{K}_A \mathbf{Q}_A^H \mathbf{f}, \quad (11)$$

where $\mathcal{K}_A = \mathbf{R}_A^{\alpha}$, and let $\alpha = \frac{2}{\pi} \cos^{-1}\left(\frac{a}{\sigma}\right) = \frac{2}{\pi} \sin^{-1}\left(\frac{b}{\sigma}\right)$.

Then we consider the scale adjacency matrix $\mathbf{S}_{A,\sigma} = \text{diag}([\sigma^{-\epsilon r_0}, \sigma^{-\epsilon r_1}, \dots, \sigma^{-\epsilon r_{N-1}}])$, where $\{r_l\}_{l=0}^{N-1}$ are the eigenvalues associated with the undilated eigenspace $\mathbf{Q}_A$, and ϵ represents the scaling adjustment parameter. The resulting scaled eigenspace $\mathbf{P}_A = [\mathbf{p}_0, \mathbf{p}_1, \dots, \mathbf{p}_{N-1}] = \mathbf{S}_{A,\sigma} \mathbf{Q}_A$ generates frequency-scaled basis vectors $\{\mathbf{p}_l\}$ that adaptively emphasize different spectral bands. The graph scale transformation for a signal $\mathbf{f}$ is expressed as [27]

$$\mathcal{S}T_{\sigma}^{(A)} \mathbf{f} = \mathbf{P}_A \mathbf{Q}_A^H \mathbf{f}, \quad (12)$$

where the scaling factor σ is computed as $\sigma = \sqrt{a^2 + b^2}$.

Finally we define a signal in a graph spectral domain as

$$\hat{s}_{\xi}^{(A)}(k) = \begin{cases} e^{-i\frac{\mu_k^2}{\xi}}, & \xi \neq 0, \\ \mathbf{1}_N, & \xi = 0 \end{cases}, \quad (13)$$

where μ_k represents the k th eigenvalue in $\mathbf{J}$, $\mathbf{1}_N$ represents an N -dimensional column vector with all elements being 1. Next, the inverse

GFT is performed to obtain its vertex domain form as $s(n) = \mathbf{U}\hat{s}_\xi(k)$. We utilize this signal as the CM term for the graph signal. Let

$$\mathbf{Y}_A = \text{diag} \left(\mathbf{V}\hat{s}_\xi^{(A)}(k) \right), \quad (14)$$

thus, defining the CM operator on the graph as [27]

$$CM_\xi^{(A)} \mathbf{f} = \mathbf{Y}_A \mathbf{f}, \quad (15)$$

where $\xi = \frac{ac+bd}{a^2+b^2}$.

The adjacency-based GLCT (A-GLCT) is defined by [27]

$$\hat{\mathbf{f}}_M^{(A)} = \mathcal{F}_M^{(A)} \mathbf{f} = \mathbf{Y}_A (\mathbf{P}_A \mathbf{Q}_A^H) (\mathbf{Q}_A \mathcal{K}_A \mathbf{Q}_A^H) \mathbf{f} = \mathbf{Y}_A \mathbf{P}_A \mathcal{K}_A \mathbf{Q}_A^H \mathbf{f}. \quad (16)$$

The corresponding eigenvalues are $\mathbf{J}_M = \text{diag}([\mu_0, \mu_1, \dots, \mu_{N-1}])$, where $\mu_l = \sigma \tilde{\mu}_l^\alpha$, $l = 0, \dots, N-1$. The inverse A-GLCT is defined as

$$\mathbf{f} = (\mathcal{F}_M^{(A)})^H = \mathbf{Q}_A \mathcal{K}_A^H \mathbf{P}_A^H \mathbf{Y}_A^H \hat{\mathbf{f}}_M^{(A)}. \quad (17)$$

We can also define the graph linear canonical adjacency operator as

$$\mathbf{W}_M^{(A)} = (\mathcal{F}_M^{(A)})^H \mathbf{J}_M \mathcal{F}_M^{(A)}. \quad (18)$$

2.3.2. Laplacian-based graph linear canonical transform

The L-GFT matrix $\mathcal{F}^{(L)}$ can be decomposed as $\mathcal{F}^{(L)} = \mathbf{Q}_L \mathbf{R}_L \mathbf{Q}_L^H$, where $\mathbf{Q}_L = [\mathbf{q}_0, \mathbf{q}_1, \dots, \mathbf{q}_{N-1}]$ and $\mathbf{R}_L = \text{diag}([r'_0, r'_1, \dots, r'_{N-1}])$. By denoting $\mathcal{K}_L = \text{diag}([\kappa_0, \kappa_1, \dots, \kappa_{N-1}]) = \mathbf{R}_L^\alpha$, $\kappa_l = (r'_l)^\alpha$, $l = 0, \dots, N-1$, the Laplacian-based graph fractional Fourier transform (L-GFrFT) is defined as [14]

$$\hat{\mathbf{f}}_a^{(L)} = \mathcal{F}_a^{(L)} \mathbf{f} = \mathbf{Q}_L \mathcal{K}_L \mathbf{Q}_L^H \mathbf{f}, \quad (19)$$

where $\alpha = \frac{2}{\pi} \cos^{-1} \left(\frac{a}{\sigma} \right) = \frac{2}{\pi} \sin^{-1} \left(\frac{b}{\sigma} \right)$.

Consider the scale matrix $\mathbf{S}_{L,\sigma} = \text{diag}([\sigma^{-\epsilon r'_0}, \sigma^{-\epsilon r'_1}, \dots, \sigma^{-\epsilon r'_{N-1}}])$, where $\{r'_l\}_{l=0}^{N-1}$ are the eigenvalues associated with the undilated eigenspace $\mathbf{Q}_L$, and ϵ represents the scaling adjustment parameter. The resulting scaled eigenspace $\mathbf{P}_L = [\mathbf{p}'_0, \mathbf{p}'_1, \dots, \mathbf{p}'_{N-1}] = \mathbf{S}_{L,\sigma} \mathbf{Q}_L$ generates frequency-scaled basis vectors $\{\mathbf{p}'_l\}$ that adaptively emphasize different spectral bands. The graph scale transform is defined as [27]

$$ST_\sigma^{(L)} \mathbf{f} = \mathbf{P}_L \mathbf{Q}_L^H \mathbf{f}, \quad (20)$$

where $\sigma = \sqrt{a^2 + b^2}$.

Finally, we define a signal in a graph spectral domain as

$$\hat{s}_\xi^{(L)}(k) = \begin{cases} e^{-i \frac{\tilde{\omega}_k^2}{\xi}}, & \xi \neq 0, \\ \mathbf{1}_N, & \xi = 0 \end{cases} \quad (21)$$

where $\mathbf{1}_N$ represents an N-dimensional column vector with all elements being 1. Next, the inverse GFT is performed to obtain its vertex domain form as $s(n) = \mathbf{U}\hat{s}_\xi^{(L)}(k)$. We utilize this signal as the CM term for the graph signal. Let

$$\mathbf{Y}_L = \text{diag} \left(\mathbf{U}\hat{s}_\xi^{(L)}(k) \right), \quad (22)$$

thus, defining the CM operator on the graph as [27]

$$CM_\xi^{(L)} \mathbf{f} = \mathbf{Y}_L \mathbf{f}, \quad (23)$$

where $\xi = \frac{ac+bd}{a^2+b^2}$.

The Laplacian-based GLCT (L-GLCT) is defined by [27]

$$\hat{\mathbf{f}}_M^{(L)}(k) = \sum_{n=0}^{N-1} \sum_{l=0}^{N-1} f(n) \epsilon_k p_l(k) \kappa_l q_l^*(n). \quad (24)$$

Its vector form is

$$\hat{\mathbf{f}}_M^{(L)} = \mathcal{F}_M^{(L)} \mathbf{f} = \mathbf{Y}_L (\mathbf{P}_L \mathbf{Q}_L^H) (\mathbf{Q}_L \mathcal{K}_L \mathbf{Q}_L^H) \mathbf{f} = \mathbf{Y}_L \mathbf{P}_L \mathcal{K}_L \mathbf{Q}_L^H \mathbf{f}. \quad (25)$$

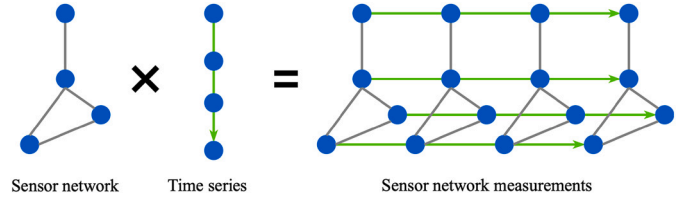


Fig. 1. Measurements of a sensor network.

The corresponding eigenvalues are $\mathbf{\Lambda}_M = \text{diag}([\lambda_0, \lambda_1, \dots, \lambda_{N-1}])$, where $\lambda_l = \sigma \tilde{\lambda}_l^\alpha$, $l = 0, \dots, N-1$. The inverse L-GLCT is defined as

$$\mathbf{f} = (\mathcal{F}_M^{(L)})^H = \mathbf{Q}_L \mathcal{K}_L^H \mathbf{P}_L^H \mathbf{Y}_L^H \hat{\mathbf{f}}_M^{(L)}. \quad (26)$$

We can also define the graph linear canonical Laplacian operator as

$$\mathcal{L}_M = (\mathcal{F}_M^{(L)})^H \mathbf{\Lambda}_M \mathcal{F}_M^{(L)}. \quad (27)$$

2.4. Cartesian product graph

Consider two graphs $\mathcal{G}_1 = (\mathcal{V}_1, \mathcal{E}_1, \mathbf{W}_1)$ and $\mathcal{G}_2 = (\mathcal{V}_2, \mathcal{E}_2, \mathbf{W}_2)$. Let $\mathcal{L}_1$ and $\mathcal{L}_2$ be the Laplacian matrices of $\mathcal{G}_1$ and $\mathcal{G}_2$, respectively. The Cartesian product graph $\mathcal{G}_1 \times \mathcal{G}_2$ is defined with the vertex set $\mathcal{V}_1 \times \mathcal{V}_2$, where $\mathcal{V}_1 = \{0, \dots, N_1 - 1\}$ and $\mathcal{V}_2 = \{0, \dots, N_2 - 1\}$ [16]. The edge set of $\mathcal{G}_1 \times \mathcal{G}_2$ is defined by the following condition

$$\{(i_1, i_2), (j_1, j_2)\} \in \mathcal{E}, \quad (28)$$

if and only if

$$(i_1 = j_1, \{i_2, j_2\} \in \mathcal{E}_2) \quad \text{or} \quad (i_2 = j_2, \{i_1, j_1\} \in \mathcal{E}_1). \quad (29)$$

The weight function w is given by

$$w((i_1, i_2), (j_1, j_2)) = \mathbf{W}_1(i_1, j_1) \delta_{i_2 j_2} + \mathbf{W}_2(i_2, j_2) \delta_{i_1 j_1}, \quad (30)$$

where δ is the delta function, i.e., $\delta_{ij} = 1$ if $i = j$ and $\delta_{ij} = 0$ otherwise.

Based on this definition, the Cartesian product graphs satisfy the following properties [25]:

- Associative: The graphs $(\mathcal{G}_1 \times \mathcal{G}_2) \times \mathcal{G}_3$ and $\mathcal{G}_1 \times (\mathcal{G}_2 \times \mathcal{G}_3)$ are naturally isomorphic.
- Commutative: The graphs $\mathcal{G}_1 \times \mathcal{G}_2$ and $\mathcal{G}_2 \times \mathcal{G}_1$ are naturally isomorphic.

Additionally, the graphs $\mathcal{G}_1$ and $\mathcal{G}_1$ are factor graphs of $\mathcal{G}_1 \times \mathcal{G}_1$. The adjacency matrix $\mathbf{W}_\times$ and the Laplacian matrix $\mathcal{L}_\times$ of $\mathcal{G}_1 \times \mathcal{G}_2$ can be expressed by those of its factor graphs as follows

$$\mathbf{W}_\times = \mathbf{W}_1 \oplus \mathbf{W}_2 = \mathbf{W}_1 \otimes \mathbf{I}_{N_2} + \mathbf{I}_{N_1} \otimes \mathbf{W}_2, \quad (31)$$

$$\mathcal{L}_\times = \mathcal{L}_1 \oplus \mathcal{L}_2 = \mathbf{I}_{N_2} \otimes \mathcal{L}_1 + \mathcal{L}_2 \otimes \mathbf{I}_{N_1}. \quad (32)$$

Here, the operator $\oplus$ represents the Kronecker sum and $\otimes$ represents the Kronecker product. Using these definitions, the eigenvalues of $\mathcal{G}_1 \times \mathcal{G}_2$ are all possible sums of the eigenvalues of $\mathcal{G}_1$ and $\mathcal{G}_2$: $\lambda_i(\mathcal{G}_1) + \lambda_j(\mathcal{G}_2)$, where $0 \leq i \leq N_1 - 1$ and $0 \leq j \leq N_2 - 1$.

We present a simple example to illustrate a Cartesian product graph. The sensor network measurements are decomposed into the Cartesian product of the sensor network and the time series, as shown in Fig. 1.

3. Multi-dimensional graph linear canonical transform

3.1. Two-dimensional graph linear canonical transform

3.1.1. Adjacency-based 2D graph linear canonical transform

Consider a Cartesian product graph $\mathcal{G}_1 \times \mathcal{G}_2$, their adjacency operators are defined as $\mathbf{W}_1$, $\mathbf{W}_2$, and the eigendecomposition (2) of $\mathbf{W}_l$ is

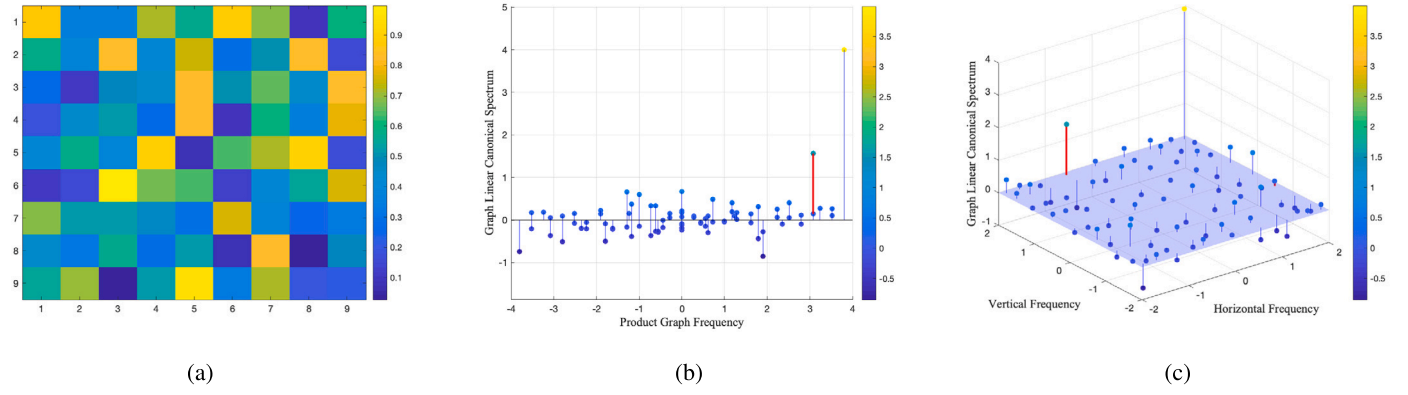


Fig. 2. The A-2D GLCT of a random 2D graph signal with parameters $\sigma = 2$, $\xi = 0$, and $\alpha = 1$. (a) The original signal. (b) The graph linear canonical spectrum. (c) The 2D graph linear canonical spectrum.

respectively $\mathbf{W}_i = \mathbf{V}_i \mathbf{J}_i \mathbf{V}_i^{-1}$, $i = 1, 2$. The eigendecomposition of adjacency operator $\mathbf{W}_x = \mathbf{W}_1 \oplus \mathbf{W}_2$ is

$$\begin{aligned} \mathbf{W}_x &= \mathbf{W}_1 \oplus \mathbf{W}_2 = \mathbf{W}_1 \otimes \mathbf{I}_{N_2} + \mathbf{I}_{N_1} \otimes \mathbf{W}_2 \\ &= (\mathbf{V}_1 \otimes \mathbf{V}_2) \left(\mathbf{J}_1 \otimes \mathbf{I}_{N_2} + \mathbf{I}_{N_1} \otimes \mathbf{J}_2 \right) (\mathbf{V}_1 \otimes \mathbf{V}_2)^{-1}. \end{aligned} \quad (33)$$

Using the above eigendecomposition, we define the adjacency-based 2D GLCT (A-2D GLCT).

Definition 1. (A-2D GLCT) Let $(\mathbf{V}_1 \otimes \mathbf{V}_2)^{-1} = \mathbf{Q}_A^\times \mathbf{R}_A^\times (\mathbf{Q}_A^\times)^H$, $\mathcal{K}_A^\times = (\mathbf{R}_A^\times)^\alpha$. Denote the scale adjacency operator $\mathbf{S}_A^\times = \sigma \mathbf{W}_x = \mathbf{Y}_A^\times \mathbf{X}_A^\times (\mathbf{Y}_A^\times)^H$, and $(\mathbf{Y}_A^\times)^H = \mathbf{P}_A^\times \Xi_A^\times (\mathbf{P}_A^\times)^H$. Let $\Upsilon_A^\times = \text{diag} \left((\mathbf{V}_1 \otimes \mathbf{V}_2) \hat{s}_\xi^{(A)}(k) \right)$, where $\hat{s}_\xi^{(A)}(k)$ is defined in (13). The A-2D GLCT of a 2D signal $\mathbf{f} : \mathcal{V} \times \mathcal{V} \rightarrow \mathbb{R}$ on a Cartesian product graph $\mathcal{G}_1 \times \mathcal{G}_2$ is

$$\hat{\mathbf{f}}_{\mathbf{M}\times}^{(A)} = \mathcal{F}_{\mathbf{M}\times}^{(A)} \mathbf{f} = \Upsilon_A^\times \mathbf{P}_A^\times \mathcal{K}_A^\times (\mathbf{Q}_A^\times)^H \mathbf{f}. \quad (34)$$

Remark 1. Define a signal on a Cartesian product graph as a “2D signal”. The 2D signal is not actual 2D data [31]. Instead, it is a 2D graph signal, which is distributed across the nodes of a 2D graph. This concept can naturally be generalized to multi-dimensional (mD) graph signals.

Since the spectrum of a 2D graph signal obtained through 1D transforms is often multi-valued at specific frequencies, and the GLCT can only represent a 1D spectrum, this method is unsuitable for 2D graph signals. However, the A-2D GLCT can address the multi-valued nature of spectral functions, as demonstrated in Example 1.

Example 1. Fig. 2 visually illustrates the limitation of 1D transforms and the advantage of the A-2D GLCT with parameters $\sigma = 2$, $\xi = 0$, and $\alpha = 1$. Fig. 2(a) displays a random 2D graph signal on a grid graph (Cartesian product of two path graphs). The x and y axes represent the nine vertices of the path graphs, and the color of each node indicates the intensity of the random signal. In Fig. 2(b), it is evident that the spectrum is double-valued at certain frequencies, with a red line highlighting an example of this double-valued spectrum. Conversely, in Fig. 2(c), all spectra are single-valued. The same red line demonstrates that the double-valued spectrum in Fig. 2(b) can be identified and separated using our new transform.

3.1.2. Laplacian-based 2D graph linear canonical transform

Consider a Cartesian product graph $\mathcal{G}_1 \times \mathcal{G}_2$, their graph Laplacian matrix are defined by $\mathcal{L}^{(i)} = \mathbf{U}_i \mathbf{A}_i \mathbf{U}_i^H$, where $i = 1, 2$. Their graph linear canonical Laplacian operators are defined as

$$\mathcal{L}_{\mathbf{M}}^{(i)} = (\mathcal{F}_{\mathbf{M}}^{(Li)})^H \mathbf{A}_{\mathbf{M}}^{(i)} \mathcal{F}_{\mathbf{M}}^{(Li)},$$

$$\mathcal{F}_{\mathbf{M}}^{(Li)} = \mathbf{Y}_{Li} \mathbf{P}_{Li} \mathcal{K}_{Li} \mathbf{Q}_{Li}^H, \quad (35)$$

$$\mathbf{A}_{\mathbf{M}}^{(i)} = \sigma (\mathbf{A}_i)^\alpha, \quad (36)$$

and $\mathbf{Y}_{Li}$, $\mathbf{P}_{Li}$, $\mathcal{K}_{Li}$, $\mathbf{Q}_{Li}$ are given by

$$\begin{aligned} \mathcal{F}^{(Li)} &= \mathbf{U}_i^H = \mathbf{Q}_{Li} \mathbf{R}_{Li} \mathbf{Q}_{Li}^H, \\ \mathbf{S}_{Li} &= \sigma \mathcal{L}^{(i)} = \mathbf{Y}_{Li} \mathbf{X}_{Li} \mathbf{Y}_{Li}^H, \quad \mathbf{Y}_{Li}^H = \mathbf{P}_{Li} \Xi_{Li} \mathbf{P}_{Li}^H, \\ \mathbf{Y}_{Li} &= \text{diag} \left(\mathbf{U}_i \hat{s}_\xi^{(Li)}(k) \right), \end{aligned} \quad (37)$$

where $\mathbf{Q}_{Li} = [\mathbf{q}_0^{(i)}, \mathbf{q}_1^{(i)}, \dots, \mathbf{q}_{N-1}^{(i)}]$, $\mathbf{R}_{Li} = \text{diag} \left([r_0^{(i)}, r_1^{(i)}, \dots, r_{N-1}^{(i)}] \right)$, $\mathcal{K}_{Li} = \text{diag} \left([\kappa_0^{(i)}, \kappa_1^{(i)}, \dots, \kappa_{N-1}^{(i)}] \right) = \mathbf{R}_{Li}^\alpha$, $\mathbf{P}_{Li} = [\mathbf{p}_0, \mathbf{p}_1, \dots, \mathbf{p}_{N-1}]$, $\Xi_{Li} = \text{diag} \left([t_0, t_1, \dots, t_{N-1}] \right)$, $\mathbf{A}_{\mathbf{M}} = \text{diag} \left([\lambda_0, \lambda_1, \dots, \lambda_{N-1}] \right)$, and $\hat{s}_\xi^{(Li)}(k)$ is defined in (21). More details see Section 2.3.2.

Therefore, for $\mathcal{G}_1 \times \mathcal{G}_2$, the Kronecker sum $\mathcal{L}_{\mathbf{M}}^\times = \mathcal{L}_{\mathbf{M}}^{(1)} \oplus \mathcal{L}_{\mathbf{M}}^{(2)}$ has an eigenvalue $\lambda_{k_1}^{(1)} + \lambda_{k_2}^{(2)}$ and the corresponding eigenfunction $\eta_{k_1}^{(1)}(n_1) = \sum_{l=0}^{N_1-1} \epsilon_{k_1}^* p_l^*(k_1) \kappa_l^*(n_1)$ and $\eta_{k_2}^{(2)}(n_2) = \sum_{l=0}^{N_2-1} \epsilon_{k_2}^* p_l^*(k_2) \kappa_l^*(n_2)$ satisfying

$$\begin{aligned} & \left(\mathcal{L}_{\mathbf{M}}^{(1)} \oplus \mathcal{L}_{\mathbf{M}}^{(2)} \right) \begin{pmatrix} \eta_{k_1}^{(1)}(0) \eta_{k_2}^{(2)}(0) \\ \eta_{k_1}^{(1)}(0) \eta_{k_2}^{(2)}(1) \\ \vdots \\ \eta_{k_1}^{(1)}(N_1-1) \eta_{k_2}^{(2)}(N_2-1) \end{pmatrix} \\ &= \left(\lambda_{k_1}^{(1)} + \lambda_{k_2}^{(2)} \right) \begin{pmatrix} \eta_{k_1}^{(1)}(0) \eta_{k_2}^{(2)}(0) \\ \eta_{k_1}^{(1)}(0) \eta_{k_2}^{(2)}(1) \\ \vdots \\ \eta_{k_1}^{(1)}(N_1-1) \eta_{k_2}^{(2)}(N_2-1) \end{pmatrix}, \end{aligned} \quad (38)$$

for any $k_1 = 0, \dots, N_1 - 1$ and $k_2 = 0, \dots, N_2 - 1$.

Using the eigenvalues and eigenfunctions of $\mathcal{L}_{\mathbf{M}}^{(1)} \oplus \mathcal{L}_{\mathbf{M}}^{(2)}$ in (38), we propose the Laplacian-based 2D GLCT (L-2D GLCT).

Definition 2. (L-2D GLCT) The L-2D GLCT of a 2D signal $\mathbf{f} : \mathcal{V} \times \mathcal{V} \rightarrow \mathbb{R}$ on a Cartesian product graph $\mathcal{G}_1 \times \mathcal{G}_2$ is

$$\hat{f}_{\mathbf{M}\times}^{(L)}(k_1, k_2) = \sum_{n_1=0}^{N_1-1} \sum_{n_2=0}^{N_2-1} f(n_1, n_2) \left(\eta_{k_1}^{(1)}(n_1) \right)^* \left(\eta_{k_2}^{(2)}(n_2) \right)^*, \quad (39)$$

where $\eta_{k_1}^{(1)}(n_1) = \sum_{l=0}^{N_1-1} \epsilon_{k_1}^* p_l^*(k_1) \kappa_l^*(n_1)$, $\eta_{k_2}^{(2)}(n_2) = \sum_{l=0}^{N_2-1} \epsilon_{k_2}^* p_l^*(k_2) \kappa_l^*(n_2)$, and $k_1 = 0, \dots, N_1 - 1$ and $k_2 = 0, \dots, N_2 - 1$. And its inverse L-2D GLCT is defined as

$$f(n_1, n_2) = \sum_{k_1=0}^{N_1-1} \sum_{k_2=0}^{N_2-1} \hat{f}_{\mathbf{M}\times}^{(L)}(k_1, k_2) \eta_{k_1}^{(1)}(n_1) \eta_{k_2}^{(2)}(n_2), \quad (40)$$

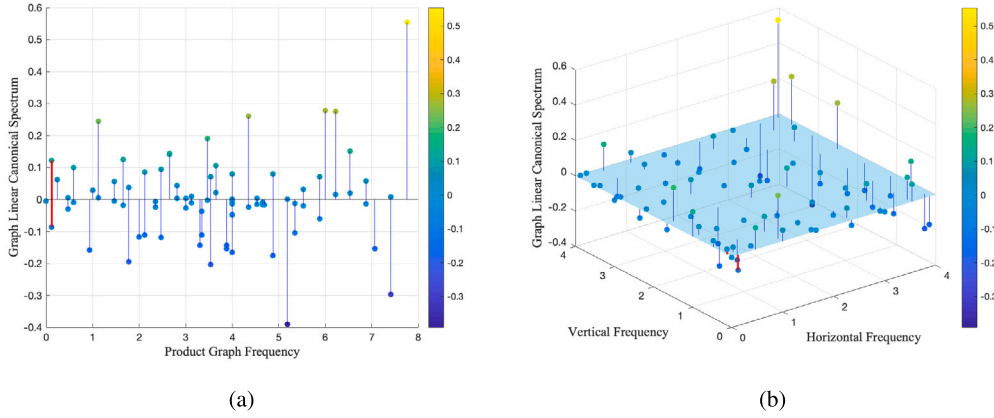


Fig. 3. The L-2D GLCT of a random 2D graph signal with parameters $\sigma = 3.04$, $\xi = -1.97$, and $\alpha = 0.11$. (a) The graph linear canonical spectrum. (b) The 2D graph linear canonical spectrum.

for $n_1 = 0, \dots, N_1 - 1$ and $n_2 = 0, \dots, N_2 - 1$.

Similarly, L-2D GLCT can also address the multi valued nature of spectral functions, as shown in Example 2.

Example 2. Continuing with the 2D random graph signal from Fig. 2(a), the L-2D GLCT with parameters $\sigma = 3.04$, $\xi = -1.97$, and $\alpha = 0.11$ is shown in Fig. 3. In Fig. 3(a), the spectrum is clearly double-valued at certain frequencies, with a red line highlighting an example of this double-valued spectrum. In contrast, Fig. 3(b) shows that all spectra are single-valued. The same red line demonstrates that the double-valued spectrum in Fig. 3(a) can be identified and separated using our new transform.

3.2. Multi-dimensional graph linear canonical transform

In this section, the multi-dimensional linear canonical transform (MGLCT), utilizing the Laplacian and adjacency matrices, is inductively defined through 2D GLCTs.

3.2.1. Adjacency-based mD graph linear canonical transform

Given a natural number m , consider m graphs $\mathcal{G}_i = (\mathcal{V}_i, \mathcal{E}_i, \mathbf{W}_i)$, $i = 1, 2, \dots, m$ with vertex set $\mathbf{V}_i = \{0, 1, \dots, N_i - 1\}$, we have a Cartesian product graph $\mathcal{G}_1 \times \mathcal{G}_2 \times \dots \times \mathcal{G}_m$. Using the same Cartesian product graph as in Section 3.1, the eigendecomposition of the adjacency operator $\mathbf{W}_i$ of graph $\mathcal{G}_i$ is $\mathbf{W}_i = \mathbf{V}_i \mathbf{J}_i \mathbf{V}_i^{-1}$. The eigendecomposition of adjacency operator $\mathbf{W}_{\times m} = \mathbf{W}_1 \oplus \mathbf{W}_2 \oplus \dots \oplus \mathbf{W}_m$ is

$$\begin{aligned} \mathbf{W}_{\times m} &= \mathbf{W}_1 \oplus \mathbf{W}_2 \oplus \dots \oplus \mathbf{W}_m \\ &= (\mathbf{V}_1 \otimes \dots \otimes \mathbf{V}_m) \left(\sum_{j=1}^m \mathbf{I}_{N_1 \dots N_{j-1}} \otimes \mathbf{J}_j \otimes \mathbf{I}_{N_{j+1} \dots N_m} \right) (\mathbf{V}_1 \otimes \dots \otimes \mathbf{V}_m)^{-1}. \end{aligned} \quad (41)$$

Using the above eigendecomposition, we define the adjacency-based mD GLCT (A-mD GLCT).

Definition 3. (A-mD GLCT) Let $(\mathbf{V}_1 \otimes \dots \otimes \mathbf{V}_m)^{-1} = \mathbf{Q}_A^{\times m} \mathbf{R}_A^{\times m} (\mathbf{Q}_A^{\times m})^H$, $\mathbf{K}_A^{\times m} = (\mathbf{R}_A^{\times m})^\alpha$. Denote the scale adjacency operator $\mathbf{S}_A^{\times m} = \sigma \mathbf{W}_{\times m} = \mathbf{Y}_A^{\times m} \mathbf{X}_A^{\times m} (\mathbf{Y}_A^{\times m})^H$, and $(\mathbf{Y}_A^{\times m})^H = \mathbf{P}_A^{\times m} \Xi_A^{\times m} (\mathbf{P}_A^{\times m})^H$. Let $\mathbf{Y}_A^{\times m} = \text{diag} \left((\mathbf{V}_1 \otimes \dots \otimes \mathbf{V}_m) \hat{s}_\xi^{(A)}(k) \right)$, where $\hat{s}_\xi^{(A)}(k)$ is defined in (13). The A-2D GLCT of a 2D signal $\mathbf{f} : \mathcal{V} \times \mathcal{V} \times \dots \times \mathcal{V} \rightarrow \mathbb{R}$ on a Cartesian product graph $\mathcal{G}_1 \times \mathcal{G}_2 \times \dots \times \mathcal{G}_m$ is

$$\hat{\mathbf{f}}_{M \times m}^{(A)} = \mathcal{F}_{M \times m}^{(A)} \mathbf{f} = \mathbf{Y}_A^{\times m} \mathbf{P}_A^{\times m} \mathbf{K}_A^{\times m} (\mathbf{Q}_A^{\times m})^H \mathbf{f}. \quad (42)$$

3.2.2. Laplacian-based mD graph linear canonical transform

Consider a Cartesian product graph $\mathcal{G}_1 \times \mathcal{G}_2 \times \dots \times \mathcal{G}_m$, their graph Laplacian matrix are defined by $\mathcal{L}^{(i)} = \mathbf{U}_i \mathbf{A}_i \mathbf{U}_i^H$, where $i = 1, 2, \dots, m$. Their graph linear canonical Laplacian operators are defined as

$$\mathcal{L}_M^{(i)} = (\mathcal{F}_M^{(Li)})^H \mathbf{A}_M^{(i)} \mathcal{F}_M^{(Li)}, \quad (43)$$

$$\mathcal{F}_M^{(Li)} = \mathbf{Y}_{Li} \mathbf{P}_{Li} \mathbf{K}_{Li} \mathbf{Q}_{Li}^H, \quad (44)$$

$$\mathbf{A}_M^{(i)} = \sigma (\mathbf{A}_i)^\alpha, \quad (45)$$

and $\mathbf{Y}_{Li}$, $\mathbf{P}_{Li}$, $\mathbf{K}_{Li}$, $\mathbf{Q}_{Li}$ are given by $\mathcal{F}^{(Li)} = \mathbf{U}_i^H = \mathbf{Q}_{Li} \mathbf{R}_{Li} \mathbf{Q}_{Li}^H$, $\mathbf{S}_{Li} = \sigma \mathcal{L}^{(i)} = \mathbf{Y}_{Li} \mathbf{X}_{Li} \mathbf{Y}_{Li}^H$, $\mathbf{Y}_{Li}^H = \mathbf{P}_{Li} \Xi_{Li} \mathbf{P}_{Li}^H$, $\mathbf{Y}_{Li} = \text{diag} \left(\mathbf{U}_i \hat{s}_\xi^{(Li)}(k) \right)$, where $\mathbf{Q}_{Li} = [\mathbf{q}_0^{(i)}, \mathbf{q}_1^{(i)}, \dots, \mathbf{q}_{N-1}^{(i)}]$, $\mathbf{R}_{Li} = \text{diag} \left([r_0^{(i)}, r_1^{(i)}, \dots, r_{N-1}^{(i)}] \right)$, $\mathbf{K}_{Li} = \text{diag} \left([\kappa_0^{(i)}, \kappa_1^{(i)}, \dots, \kappa_{N-1}^{(i)}] \right) = \mathbf{R}_{Li}^\alpha$, $\mathbf{P}_{Li} = [\mathbf{p}_0, \mathbf{p}_1, \dots, \mathbf{p}_{N-1}]$, $\Xi_{Li} = \text{diag} \left([l_0, l_1, \dots, l_{N-1}] \right)$, $\mathbf{A}_M = \text{diag} \left([\lambda_0, \lambda_1, \dots, \lambda_{N-1}] \right)$, and $\hat{s}_\xi^{(Li)}(k)$ is defined in (21). More details see Section 2.3.2.

Because the Cartesian product graph $\mathcal{G}_1 \times \mathcal{G}_2 \times \dots \times \mathcal{G}_m = ((\mathcal{G}_1 \times \mathcal{G}_2) \times \dots \times \mathcal{G}_m)$, the Kronecker sum $\mathcal{L}_M^{\times m} = \mathcal{L}_M^{(1)} \oplus \mathcal{L}_M^{(2)} \oplus \dots \oplus \mathcal{L}_M^{(m)}$ has an eigenvalue $\lambda_{k_1}^{(1)} + \lambda_{k_2}^{(2)} + \dots + \lambda_{k_m}^{(m)}$ and the corresponding eigenfunction $\eta_{k_i}^{(i)}(n_i) = \sum_{l=0}^{N_i-1} \epsilon_{k_i}^* p_l^*(k_i) \kappa_l^* q_l(n_i)$ satisfying

$$\begin{aligned} & \left(\mathcal{L}_M^{(1)} \oplus \mathcal{L}_M^{(2)} \oplus \dots \oplus \mathcal{L}_M^{(m)} \right) \begin{pmatrix} \eta_{k_1}^{(1)}(0) \eta_{k_2}^{(2)}(0) \dots \eta_{k_m}^{(m)}(0) \\ \eta_{k_1}^{(1)}(0) \eta_{k_2}^{(2)}(1) \dots \eta_{k_m}^{(m)}(1) \\ \vdots \\ \eta_{k_1}^{(1)}(N_1-1) \eta_{k_2}^{(2)}(N_2-1) \dots \eta_{k_m}^{(m)}(N_m-1) \end{pmatrix} \\ &= \left(\lambda_{k_1}^{(1)} + \lambda_{k_2}^{(2)} + \dots + \lambda_{k_m}^{(m)} \right) \begin{pmatrix} \eta_{k_1}^{(1)}(0) \eta_{k_2}^{(2)}(0) \dots \eta_{k_m}^{(m)}(0) \\ \eta_{k_1}^{(1)}(0) \eta_{k_2}^{(2)}(1) \dots \eta_{k_m}^{(m)}(1) \\ \vdots \\ \eta_{k_1}^{(1)}(N_1-1) \eta_{k_2}^{(2)}(N_2-1) \dots \eta_{k_m}^{(m)}(N_m-1) \end{pmatrix}, \end{aligned} \quad (46)$$

for any $k_i = 0, \dots, N_i - 1$.

Definition 4. (L-mD GLCT) The L-mD GLCT of a mD signal $\mathbf{f} : \mathcal{V} \times \mathcal{V} \times \dots \times \mathcal{V} \rightarrow \mathbb{R}$ on a Cartesian product graph $\mathcal{G}_1 \times \mathcal{G}_2 \times \dots \times \mathcal{G}_m$ is

$$\begin{aligned} & \hat{f}_{M \times m}^{(L)}(k_1, k_2, \dots, k_m) \\ &= \sum_{n_1=0}^{N_1-1} \sum_{n_2=0}^{N_2-1} \dots \sum_{n_m=0}^{N_m-1} f(n_1, n_2, \dots, n_m) \left(\eta_{k_1}^{(1)}(n_1) \right)^* \dots \left(\eta_{k_m}^{(m)}(n_m) \right)^*, \end{aligned} \quad (47)$$

where $\eta_{k_i}^{(i)}(n_i) = \sum_{l=0}^{N_i-1} \epsilon_{k_i}^* p_l^*(k_i) \kappa_l^* q_l(n_i)$, and $k_i = 0, \dots, N_i - 1, i = 1, 2, \dots, m$. And its inverse L-mD GLCT is defined as

$$f(n_1, n_2, \dots, n_m) = \sum_{k_1=0}^{N_1-1} \sum_{k_2=0}^{N_2-1} \dots \sum_{k_m=0}^{N_m-1} \hat{f}_{M \times m}^{(L)}(k_1, k_2, \dots, k_m) \eta_{k_1}^{(1)}(n_1) \dots \eta_{k_m}^{(m)}(n_m), \quad (48)$$

for $n_i = 0, \dots, N_i - 1, i = 1, 2, \dots, m$.

3.2.3. Discussions

In graph signal processing, two classical forms of the GLCT are commonly used: one based on the Laplacian matrix and the other on the adjacency matrix. For future applications, we extend these classical definitions to the multidimensional case. The choice of method depends on specific application requirements. For instance, if the adjacency matrix of the graph structure is known, the A-mD GLCT can be utilized. Conversely, if the graph is undirected, the L-mD GLCT is often more convenient. This flexibility allows for the selection of the most appropriate method based on the characteristics of the graph and the nature of the analysis.

As demonstrated in Examples 1 and 2, the 2D GLCTs effectively address the multi-valued nature of spectral functions, highlighting the advantages of our multidimensional LCT over traditional GLCT for signals on Cartesian product graphs. Another significant advantage is the reduction in computational time. Cartesian product graphs can effectively represent multi-domain data, and using two or more factor graphs with fewer nodes to replace the original graph can significantly reduce computational costs. For instance, consider a Cartesian product graph $\mathcal{G}_1 \times \mathcal{G}_2$. The L-2D GLCT requires $O(N_1^3 + N_2^3)$ time, whereas the traditional GLCT on the same graph costs $O(N_1^3 N_2^3)$. Additionally, when using factor graphs $\mathcal{G}_1$ and $\mathcal{G}_2$ to compute the eigendecomposition of the graph fractional Laplacian, the computational time is further reduced compared to working with the original graph. This efficiency makes the L-2D GLCT particularly advantageous for large-scale graph signal processing.

4. Application

To demonstrate the effectiveness of the mD GLCT framework, we consider its application to data compression using real-world data from the national oceanic and atmospheric administration (NOAA)'s national centers for environmental information (NCEI). We randomly selected 250 weather observation stations and extracted their temperature records for the year 2023. The underlying structure of the data can be divided into two aspects: one based on the timeline of the days in a year, and the other based on the spatial distribution of the various temperature measurement stations. The data is publicly available at <https://www.ncei.noaa.gov/data/global-summary-of-the-day/access/2023/>.

First, we construct a natural path graph of the 365-day temperature records from the temperature observation stations, denoted as $\mathcal{G}_1 = (\mathcal{V}_1, \mathcal{E}_1, \mathbf{W}_1)$. Here, $\mathcal{V}_1 = \{v_0, v_1, \dots, v_{364}\}$ represents the set of vertices corresponding to each day, and $\mathcal{E}_1 = \{(v_i, v_{i+1}) \mid 0 \leq i < 364\}$ represents the set of edges connecting consecutive days. The adjacency matrix $\mathbf{W}_1$ is defined as follows

$$\mathbf{W}_1(i, j) = \begin{cases} 1, & \text{if } |i - j| = 1 \\ 0, & \text{otherwise} \end{cases}. \quad (49)$$

This path graph structure captures the sequential nature of daily temperature observations, with each node connected to its immediate predecessor and successor.

Next, we construct a graph model for the stations using the k-nearest neighbors (kNN) method as described in [48], denoted as $\mathcal{G}_2 = (\mathcal{V}_2, \mathcal{E}_2, \mathbf{W}_2)$. For each pair of stations, we calculate the spherical distance using their latitude and longitude coordinates. Specifically, let (θ_i, ϕ_i) and (θ_j, ϕ_j) represent the coordinates of two stations i and j in radians, where $i, j \in \{0, 1, \dots, 249\}$. The spherical distance is defined as

$$d_{ij} = \arccos(\cos(\theta_i)\cos(\theta_j)\cos(\phi_i - \phi_j) + \sin(\theta_i)\sin(\theta_j)). \quad (50)$$

Based on these spherical distances, we identify the neighbors for each station. The kNN graph $\mathcal{G}_2$ is constructed by setting each node as a station, denoted as $\mathcal{V}_2 = \{v'_0, v'_1, \dots, v'_{249}\}$. The edge set $\mathcal{E}_2$ consists of edges connecting each vertex to its k-nearest neighbors. For each vertex v'_i ($0 \leq i \leq 249$), we connect it to the k vertices closest to it based on the spherical distance. The weight matrix $\mathbf{W}_2$ is defined by applying the spherical distances to the Gaussian kernel. For two nodes i and j connected by an edge, the weight w_{ij} is given by

$$\mathbf{W}_2(i, j) = \frac{\exp\left(-\frac{d_{ij}^2}{\epsilon^2}\right)}{\sqrt{\sum_{k \sim i} \exp\left(-\frac{d_{ik}^2}{\epsilon^2}\right)} \sqrt{\sum_{k \sim j} \exp\left(-\frac{d_{jk}^2}{\epsilon^2}\right)}}, \quad (51)$$

where $\sim$ indicates that two nodes are connected by an edge, and ϵ^2 is a preset parameter. It is important to note that the Earth's radius $R = 6,357$ km is omitted in the computation of the spherical distance to avoid significant precision errors when using the Gaussian kernel. Fig. 4 illustrates the kNN graph with $k = 5$, demonstrating the connections between stations based on their nearest neighbors.

Finally, we construct a product graph $\mathcal{G}_1 \times \mathcal{G}_2$ based on the path graph $\mathcal{G}_1$ of temperature records and the kNN graph $\mathcal{G}_2$ of temperature observation stations. This product graph captures the temporal and spatial relationships inherent in the data. The temperature data is then represented as a 2D graph signal on the product graph, denoted by $\mathbf{f}$, where each element $f(t, i)$ indicates the temperature at station i on day t . This representation allows for a comprehensive analysis of the temperature variations across both time and space.

Without loss of generality, the following explanation will focus on the L-2D GLCT. The compression process proceeds as follows:

- Compute the L-2D GLCT of $f(t, i)$ using (39), resulting in $\hat{f}_{M \times M}^{(L)}(k_1, k_2)$.
- Sort the coefficients $\hat{f}_{M \times M}^{(L)}(k_1, k_2)$ by their absolute values.
- Retain the largest μ fraction of the coefficients, where $0 < \mu < 1$, and set the remaining coefficients to zero.
- Perform the inverse L-2D GLCT on the modified coefficients using (40), resulting in $f_{cp}(t, i)$.

Fig. 5 displays a comparison between a 15×15 block of the original temperature data and the compressed temperature data. This highlights the efficacy of the compression technique.

To further demonstrate the effectiveness of the compression, we propose using the normalized root mean square (NRMS) and peak signal-to-noise ratio (PSNR) to quantify the error between the compressed signal $f_{cp}(t, i)$ and the original signal $f(t, i)$. The NRMS is defined as follows

$$\text{NRMS} = \frac{\sqrt{\sum_{t \in \mathcal{G}_1, i \in \mathcal{G}_2} (f(t, i) - f_{cp}(t, i))^2}}{\sqrt{\sum_{t \in \mathcal{G}_1, i \in \mathcal{G}_2} (f(t, i) - \bar{f})^2}}, \quad (52)$$

where $\bar{f} = \frac{1}{|\mathcal{G}_1||\mathcal{G}_2|} \sum_{t \in \mathcal{G}_1, i \in \mathcal{G}_2} f(t, i)$ represents the mean of the original signal. By subtracting the mean in the denominator, the influence of shifting is eliminated, leading to a more accurate assessment of compression quality. A smaller NRMS indicates better compression performance. The PSNR is given by

$$\text{PSNR} = 10 \log_{10} \left(\frac{\max_{t \in \mathcal{G}_1, i \in \mathcal{G}_2} |f_{cp}(t, i)|^2}{\frac{1}{|\mathcal{G}_1||\mathcal{G}_2|} \sum_{t \in \mathcal{G}_1, i \in \mathcal{G}_2} |f(t, i) - f_{cp}(t, i)|^2} \right). \quad (53)$$

Higher PSNR values indicate that the compressed signal retains more information from the original signal, thus reflecting higher fidelity. These metrics, NRMS and PSNR, collectively provide a comprehensive evaluation

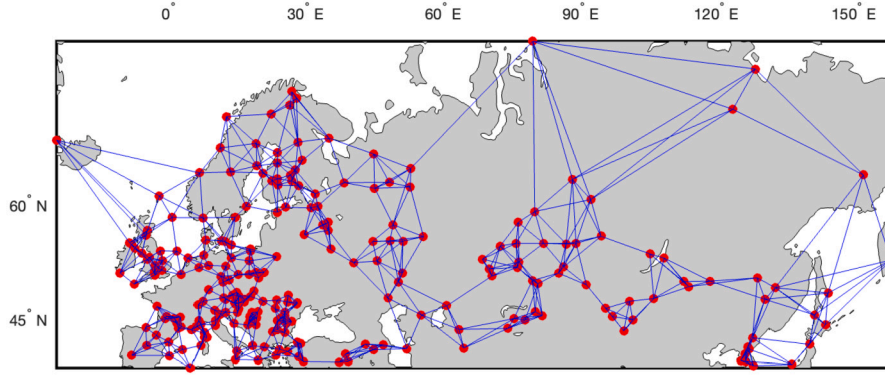


Fig. 4. Measurements of a sensor network.

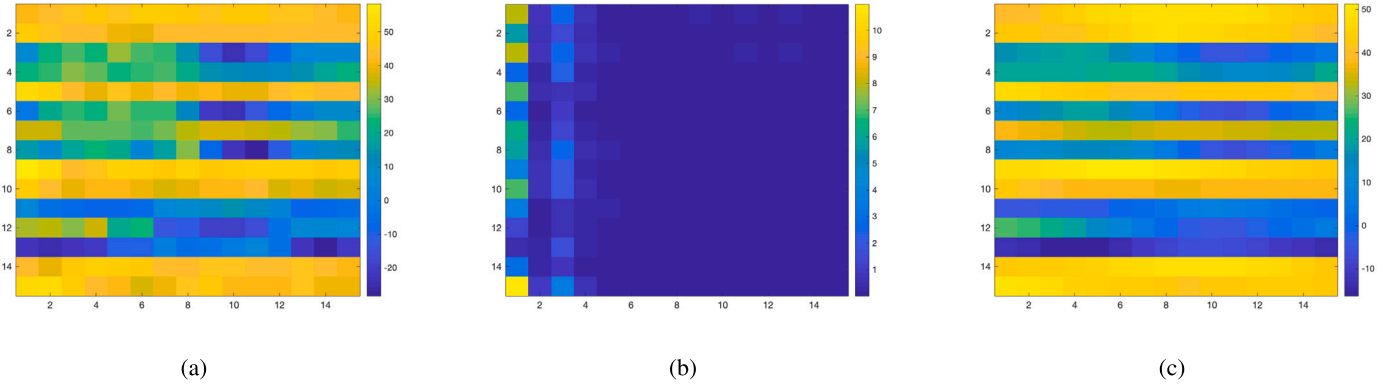
Fig. 5. An illustration of the compression process with $\sigma = 0.5$, $\xi = 0$, $\alpha = 1$ and $\mu = 0.05$. (a) The original signal. (b) The graph linear canonical spectrum. (c) Compressed signal.

Table 1

The NRMS and PSNR values for different compression parameters.

	$\mu = 0.01$	$\mu = 0.05$	$\mu = 0.1$	$\mu = 0.15$	$\mu = 0.2$	$\mu = 0.25$	$\mu = 0.3$	$\mu = 0.35$
NRMS (%)	33.19	24.34	19.54	16.44	14.13	12.31	10.80	9.50
PSNR (dB)	83.71	87.13	89.37	91.23	92.89	94.27	95.54	96.87

ation of the compression effectiveness, ensuring both the minimization of error and the maximization of retained signal quality.

Three experiments were conducted: first, Example 3 was used to examine the impact of different compression coefficients; second, the influence of different graph linear canonical parameters was investigated using Example 4; third, Example 5 was employed to compare the effectiveness of existing methods. The results demonstrate that the method proposed in this article can effectively apply to data compression, achieving better results due to the flexibility of the parameters.

Example 3. Let the graph linear canonical parameters be $\sigma = 0.5$, $\xi = 0$, and $\alpha = 1$. The compression results under various compression coefficients μ are summarized in Table 1. The data indicates that even with very small values of μ , the compression process introduces only a minimal NRMS. This demonstrates the efficacy of the method in preserving the integrity of the original signal despite significant data reduction.

Example 4. With a compression coefficient of $\mu = 0.05$ and the graph linear canonical parameter $\xi = 0$, we explore various combinations of $\alpha \in [0.5, 1]$ and $\sigma \in [1, 2]$. The results for NRMS are depicted in Fig. 6(a), while those for PSNR are shown in Fig. 6(b). Optimal performance in terms of NRMS is achieved at $\alpha = 0.9$ and $\sigma = 1.4$, whereas the best PSNR results are observed at $\alpha = 0.9$ and $\sigma = 2$. The differing optimal parameters for each evaluation metric highlight the importance of adjusting

parameters according to specific experimental contexts to achieve superior compression outcomes.

Example 5. Based on the results presented in Example 4, a further comparison was conducted with existing methods. Specifically, the L-2D GFrFT ($\alpha = 0.85$), as defined in [25], and the L-2D GLCT, as defined in [32], were selected for comparison. It should be noted that the mD GLCT defined in [32] differs from the method proposed in this study, particularly in the definitions applied during the graph scale and graph CM stages. Moreover, the optimal parameters for the L-2D GLCT were determined as $\alpha = 0.95$ and $\sigma = 1.1$. The comparison results are summarized in Table 2.

5. Conclusion

This article proposes a method for processing multi-dimensional (mD) graph signals based on graph linear canonical transforms (GLCT) and Cartesian product graphs. Firstly, we define two-dimensional (2D) GLCTs based on the adjacency matrix and Laplacian matrix to address the limitations of one-dimensional transforms in capturing the bivariate spectra. Then, we extend the 2D method to the mD domain to develop mD GLCTs. Finally, we apply the 2D GLCT based on Laplacian matrix to data compression, with experimental results demonstrating that this method can significantly improve data compression efficiency, offering a new solution for the efficient storage and transmission of graph data.

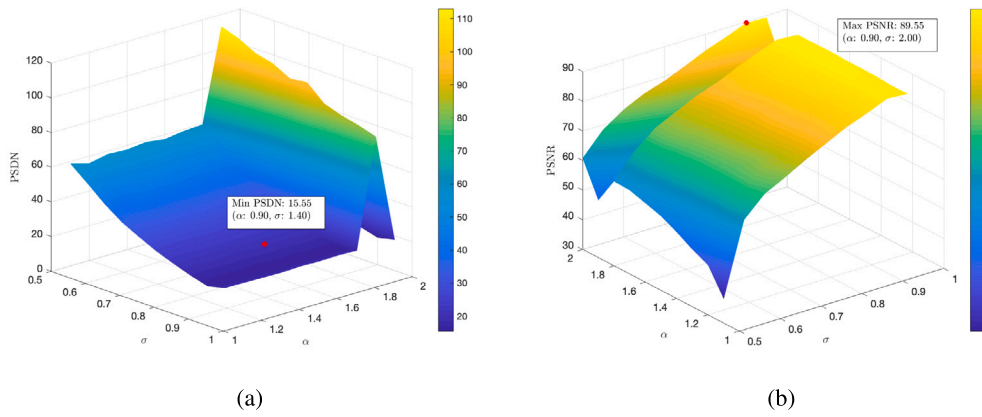


Fig. 6. The NRMS and PSNR values for different graph linear canonical parameters. (a) The NRMS. (b) The PSNR.

Table 2

NRMS and PSNR values for different methods.

Method	NRMS (%)	PSNR (dB)
L-2D GFrFT in [25]	25.53	86.80
L-2D GLCT in [32]	18.91	87.71
L-2D GLCT in this study	15.55	89.55

The parameter flexibility of GLCT enhances its potential for practical applications. The optimization of parameters will also be a major focus of our future research.

CRedit authorship contribution statement

Jian-Yi Chen: Conceptualization, Writing – original draft. **Bing-Zhao Li:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

I have shared the link to my data/code at the Attach File step.

References

- [1] F. Gama, E. Isufi, G. Leus, A. Ribeiro, Graphs, convolutions, and neural networks: from graph filters to graph neural networks, *IEEE Signal Process. Mag.* 37 (6) (2020) 128–138.
- [2] J. Miettinen, E. Nitzan, S.A. Vorobyov, E. Ollila, Graph signal processing meets blind source separation, *IEEE Trans. Signal Process.* 69 (2021) 2585–2599.
- [3] A. Sandryhaila, J.M. Moura, Big data analysis with signal processing on graphs: representation and processing of massive data sets with irregular structure, *IEEE Signal Process. Mag.* 31 (5) (2014) 80–90.
- [4] A. Sandryhaila, J.M. Moura, Discrete signal processing on graphs, *IEEE Trans. Signal Process.* 61 (7) (2013) 1644–1656.
- [5] A. Ortega, P. Frossard, J. Kovačević, J.M. Moura, P. Vanderghenst, Graph signal processing: overview, challenges, and applications, *Proc. IEEE* 106 (5) (2018) 808–828.
- [6] D.I. Shuman, Localized spectral graph filter frames: a unifying framework, survey of design considerations, and numerical comparison, *IEEE Signal Process. Mag.* 37 (6) (2020) 43–63.
- [7] Z. Xiao, H. Fang, X. Wang, Distributed nonlinear polynomial graph filter and its output graph spectrum: filter analysis and design, *IEEE Trans. Signal Process.* 69 (2021) 1725–1739.
- [8] S. Chen, A. Sandryhaila, J.M. Moura, J. Kovačević, Signal recovery on graphs: variation minimization, *IEEE Trans. Signal Process.* 63 (17) (2015) 4609–4624.
- [9] S. Rey Escudero, F.J.I. Garcia, C. Cabrera, A.G. Marques, Sampling and reconstruction of diffused sparse graph signals from successive local aggregations, *IEEE Signal Process. Lett.* 26 (8) (2019) 1142–1146.
- [10] D.K. Hammond, P. Vanderghenst, R. Gribonval, Wavelets on graphs via spectral graph theory, *Appl. Comput. Harmon. Anal.* 30 (2) (2011) 129–150.
- [11] D.I. Shuman, B. Ricaud, P. Vanderghenst, Vertex-frequency analysis on graphs, *Appl. Comput. Harmon. Anal.* 40 (2) (2016) 260–291.
- [12] Y.Q. Wang, B.Z. Li, Q.Y. Cheng, The fractional Fourier transform on graphs, in: 2017 Asia-Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC), IEEE, 2017, pp. 105–110.
- [13] J. Wu, F. Wu, Q. Yang, Y. Zhang, X. Liu, Y. Kong, L. Senhadji, H. Shu, Fractional spectral graph wavelets and their applications, *Math. Probl. Eng.* 2020 (2020) 1–18.
- [14] F.J. Yan, B.Z. Li, Windowed fractional Fourier transform on graphs: properties and fast algorithm, *Digit. Signal Process.* 118 (2021) 103210.
- [15] Y. Zhang, B.Z. Li, Discrete linear canonical transform on graphs, *Digit. Signal Process.* 135 (2023) 103934.
- [16] T. Kurokawa, T. Oki, H. Nagao, Multi-dimensional graph Fourier transform, *arXiv preprint, arXiv:1712.07811*.
- [17] A. Natali, E. Isufi, G. Leus, Forecasting multi-dimensional processes over graphs, in: ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), IEEE, 2020, pp. 5575–5579.
- [18] D.I. Shuman, S.K. Narang, P. Frossard, A. Ortega, P. Vanderghenst, The emerging field of signal processing on graphs: extending high-dimensional data analysis to networks and other irregular domains, *IEEE Signal Process. Mag.* 30 (3) (2013) 83–98.
- [19] D.K. Hammond, P. Vanderghenst, R. Gribonval, The spectral graph wavelet transform: fundamental theory and fast computation, in: *Vertex-Frequency Analysis of Graph Signals*, Springer, 2018, pp. 141–175.
- [20] X.W. Zheng, Y.Y. Tang, J.T. Zhou, H.-L. Yuan, Y.-L. Wang, L.-N. Yang, J.-J. Pan, Multi-windowed graph Fourier frames, in: 2016 International Conference on Machine Learning and Cybernetics (ICMLC), vol. 2, IEEE, 2016, pp. 1042–1048.
- [21] I. Jestrović, J.L. Coyle, E. Sejdić, A fast algorithm for vertex-frequency representations of signals on graphs, *Signal Process.* 131 (2017) 483–491.
- [22] Y.C. Gan, J.Y. Chen, B.Z. Li, The windowed two-dimensional graph fractional Fourier transform, *Digit. Signal Process.* 162 (2025) 105191.
- [23] G. Du, J. Su, L. Zhang, K. Su, X. Wang, S. Teng, P.X. Liu, A multi-dimensional graph convolution network for EEG emotion recognition, *IEEE Trans. Instrum. Meas.* 71 (2022) 1–11.
- [24] M. Neumeier, A. Tollkühn, M. Botsch, W. Utschick, A multidimensional graph Fourier transformation neural network for vehicle trajectory prediction, in: 2022 IEEE 25th International Conference on Intelligent Transportation Systems (ITSC), IEEE, 2022, pp. 687–694.
- [25] F.J. Yan, B.Z. Li, Multi-dimensional graph fractional Fourier transform and its application to data compression, *Digit. Signal Process.* 129 (2022) 103683.
- [26] Y. Zhang, Discrete linear canonical transform on graphs: fast sampling set selection method, in: 2024 6th International Conference on Image, Video and Signal Processing, 2024, pp. 162–170.
- [27] J.Y. Chen, Y. Zhang, B.Z. Li, Graph linear canonical transform: definition, vertex-frequency analysis and filter design, *IEEE Trans. Signal Process.* 72 (2024) 5691–5707.
- [28] J.J. Healy, M.A. Kutay, H.M. Ozaktas, J.T. Sheridan, *Linear Canonical Transforms: Theory and Applications*, vol. 198, Springer, 2016.
- [29] S.C. Pei, Y.C. Lai, Discrete linear canonical transforms based on dilated Hermite functions, *JOSA A* 28 (8) (2011) 1695–1708.

- [30] S.C. Pei, J.J. Ding, Eigenfunctions of linear canonical transform, *IEEE Trans. Signal Process.* 50 (1) (2002) 11–26.
- [31] A. Natali, E. Isufi, G. Leus, Forecasting multi-dimensional processes over graphs, in: *ICASSP 2020 - 2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2020, pp. 5575–5579.
- [32] N. Li, Z. Zhang, J. Han, Y. Chen, C. Cao, Multi-dimensional graph linear canonical transform, *arXiv preprint*, arXiv:2407.18931.

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